

COMPUTER SCIENCE TRIPOS Part IB – 2022 – Paper 6

9 Logic and Proof (lp15)

propositional
logic, BDDs,
DPLL

- (a) You have to write a program to classify given propositional formulas as unsatisfiable, valid or neither. You have the option of using either BDDs or DPLL for this. For each option, sketch how to carry out the classification and comment briefly on its advantages (e.g. from the standpoint of performance or space) compared with the alternative. [4 marks]

Answer: Converting a formula to a BDD immediately performs the classification: a BDD of 0 means unsatisfiable; a BDD of 1 means valid; any nontrivial BDD (with paths to both 0 and 1) means neither. With DPLL, converting the formula to clauses without negation and deriving the empty clause means that the formula is unsatisfiable. Negating the formula first and then deriving the empty clause means that the formula is valid. If the empty clause is not derived in either case then the formula is neither.

A drawback of BDDs is that they can be expected to perform badly for many formulas, because they create a representation of the entire truth table while we need only a tiny part of the truth table to make the decision. On the other hand, BDDs can handle $\leftrightarrow$ and exclusive-or without expanding them out, so presumably much more efficiently than DPLL. The other drawback of DPLL is that in general two runs are needed in the most common “neither” case.

A variety of answers are acceptable provided they rely on correct observations, such as the potentially exponential cost of conversion into clause form. It is even conceivable that a single DPLL call could perform the full classification, since conversion into CNF already detects tautologies.

propositional
logic, DPLL

- (b) It is possible to modify DPLL such that, whenever the empty clause is encountered within the search, a subset of the current variable assignments are identified that contradict the original set of clauses. These assignments are then used to create a new clause. For example, if P true, Q false, S false contradict the original clauses, then the new clause $\{\neg P, Q, S\}$ is added to the others.

- (i) In what sense is adding this clause logically correct? [2 marks]

Answer: If those variable assignments are contradictory (wrt the given clauses) then at least one of them must be false, which is what the new clause asserts. So, logically, the new clause is redundant and doesn't affect the set of models.

- (ii) How is this new clause likely to affect the operation of DPLL? [2 marks]

Answer: The same combination of variable assignments will be detected immediately. For example, if P true and Q false are assigned then $\{\neg P, Q, S\}$ will assert S true by unit propagation. Without the clause, much computation might have been necessary to refute S false.

first-order Logic,
tableau calculus

- (c) For each of the following formulas, either exhibit a formal proof in the free-variable tableau calculus, or exhibit a falsifying interpretation (a is a constant):

- (i) $[\neg P(a) \wedge \forall x (\neg P(f(x)) \rightarrow P(x))] \rightarrow \exists x (P(x) \wedge \neg P(f(x)))$ [6 marks]

Answer: After negating, conversion to NFF and Skolemizing, we get the sequent

$$\neg P(a) \wedge \forall x (P(f(x)) \vee P(x)) \wedge \forall x (\neg P(x) \vee P(f(x))) \Rightarrow$$

However, proof attempts lead nowhere. It's not hard to see that the left-side formula is true (and thus the original formula is false) in any model where $P(f(x))$ holds for all x while $P(a)$ is false. So for example let the universe be the set of natural numbers with $P(x)$ iff $x \neq 0$, $a = 0$ and $f(x) = x + 1$.

(ii) $(\forall x P(x)) \wedge (\forall x Q(x)) \vee \exists x (P(x) \rightarrow \neg Q(x))$ [6 marks]

Answer: After negating, conversion to NFF and Skolemizing, we get the sequent

$$(\neg P(a) \vee \neg Q(b)) \wedge (\forall x (P(x) \wedge Q(x))) \Rightarrow$$

Now (working upwards) we get by $\wedge I$

$$(\neg P(a) \vee \neg Q(b)), \forall x (P(x) \wedge Q(x)) \Rightarrow$$

Then by $\vee I$

$$\neg P(a), \forall x (P(x) \wedge Q(x)) \Rightarrow \quad \text{and similarly} \quad \neg Q(b), \forall x (P(x) \wedge Q(x)) \Rightarrow$$

Then by $\forall I$

$$\neg P(a), P(x) \wedge Q(x) \Rightarrow$$

Then by $\wedge I$

$$\neg P(a), P(x), Q(x) \Rightarrow$$

and this branch closes with $x = a$. With the other branch we get

$$\neg Q(b), P(y) \wedge Q(y) \Rightarrow$$

and then

$$\neg Q(b), P(y), Q(y) \Rightarrow$$

and this branch closes with $y = b$.
